MACHINE LEARNING PROJECT

Bipolar Disorder Detection

A Hybrid SMOTE–GMM–XGBoost Framework

Clinical symptom features • Imbalanced classification • Model evaluation

SMOTE
Balance training data

—

GMM
Add latent features

—

XGBoost
Predict BD label

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1. Introduction

Clinical problem

- Bipolar disorder (BD) is a chronic psychiatric illness with mania, hypomania, depression, or mixed episodes.
- Symptoms overlap unipolar depression, anxiety, PTSD, and psychosis, so misdiagnosis and delayed treatment are common.
- Standard care still depends on interviews and retrospective self-report, which miss early or atypical presentations.

Study setting

- 3,753 DSM-5 clinical records (Kaggle mental-illness table).
- 53 psychometric / symptom features; binary BD vs. non-BD.
- Goal: a tabular, interpretable screening aid — not a replacement for DSM-5 / ICD-11 diagnosis.

Why existing ML pipelines fall short

Class imbalance

Minority presentations are under-sampled. Classifiers then miss true BD cases — the error that screening cannot afford.

Hidden heterogeneity

Patients with the same label do not share one symptom profile. Raw features ignore latent subgroups.

Wrong data modality

CNNs and transformers shine on images or EEG. This problem is structured clinical table data.

Black-box risk

Psychiatric tools need feature-level reasons. Gradient-boosted trees keep accuracy and inspectability.

2. Proposed idea

SMOTE, GMM, and XGBoost are wired as a single leakage-safe flow. Oversampling happens only on training folds; GMM is a feature augmenter (not a separate clustering classifier); XGBoost is trained on original symptoms plus cluster posteriors.

1 SMOTE (train only)

For minority sample x_i and neighbour x_{nn}:

$$x_{new} = x_i + \delta (x_{nn} - x_i)$$
$$\delta \sim \text{Uniform}(0, 1)$$

Balances classes before clustering so majority Gaussian components cannot hide minority structure. The test set is never touched.

2 GMM latent features

$$p(x) = \Sigma_k \pi_k \; N(x \mid \mu_k, \Sigma_k)$$

K = 4 Gaussians. Soft membership probabilities become four extra features that encode hidden BD subtypes with overlapping symptoms — the paper's main representational claim.

3 Basic XGBoost

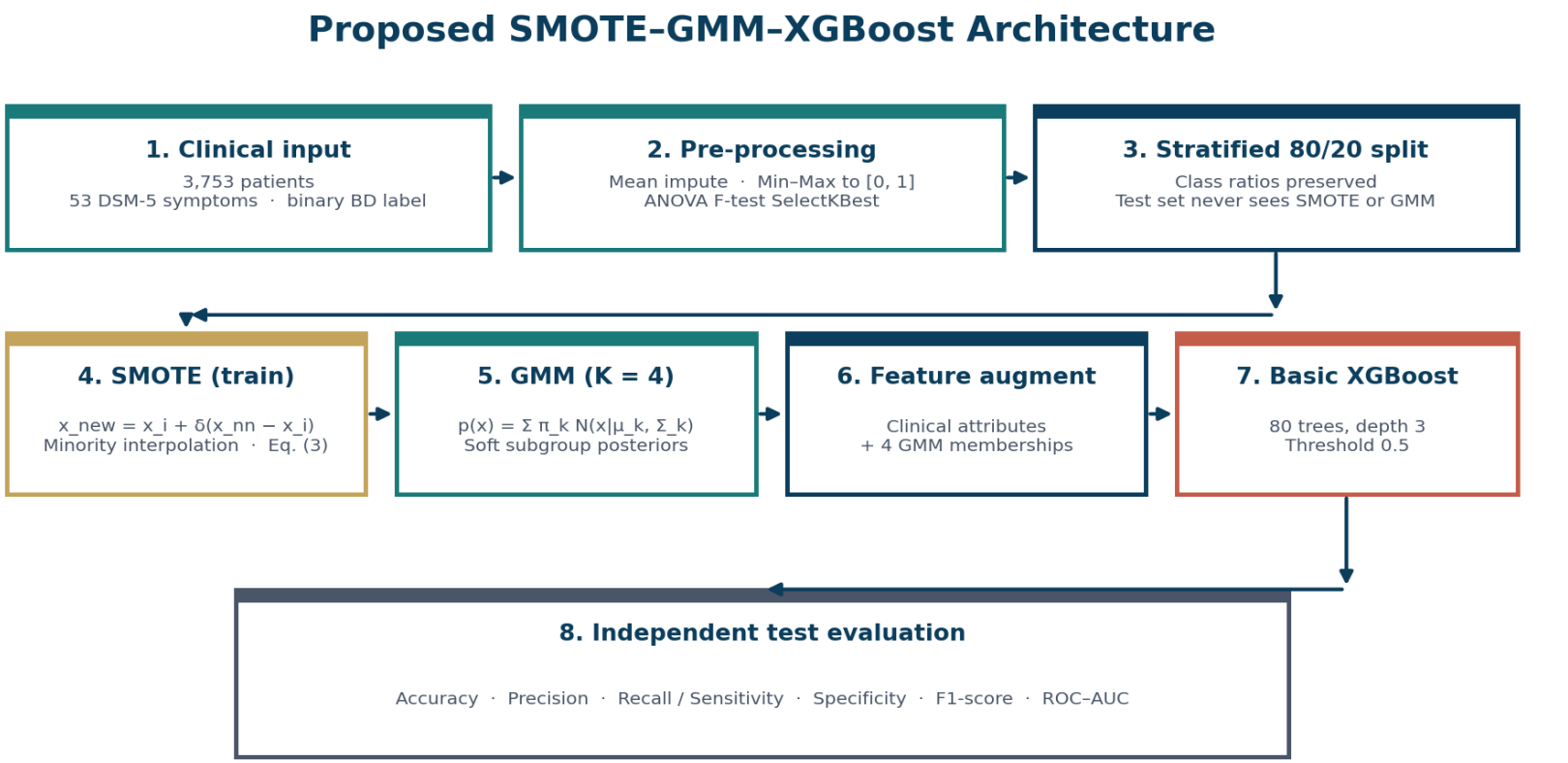
$$\hat{y}_i = \Sigma_m f_m(x_i)$$

80 shallow trees (depth 3), learning rate 0.1. No grid search — a small, regularised booster that is easy to explain. Decision threshold = 0.5.

What is new vs. prior SMOTE-only or GMM-only hybrids

GMM is not used to split the data and train one model per cluster. Posterior responsibilities are concatenated as latent variables. SMOTE runs first so subgroup discovery is not majority-biased. The boosted tree then sees both DSM-5 symptoms and cluster membership — the combination the authors argue is missing from loosely coupled earlier work.

3. Proposed architecture



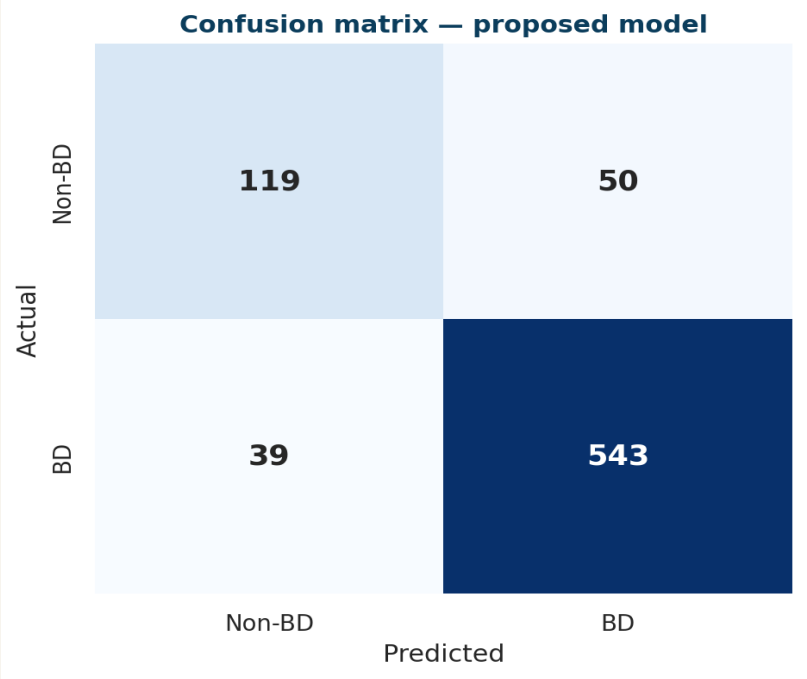
Python implementation

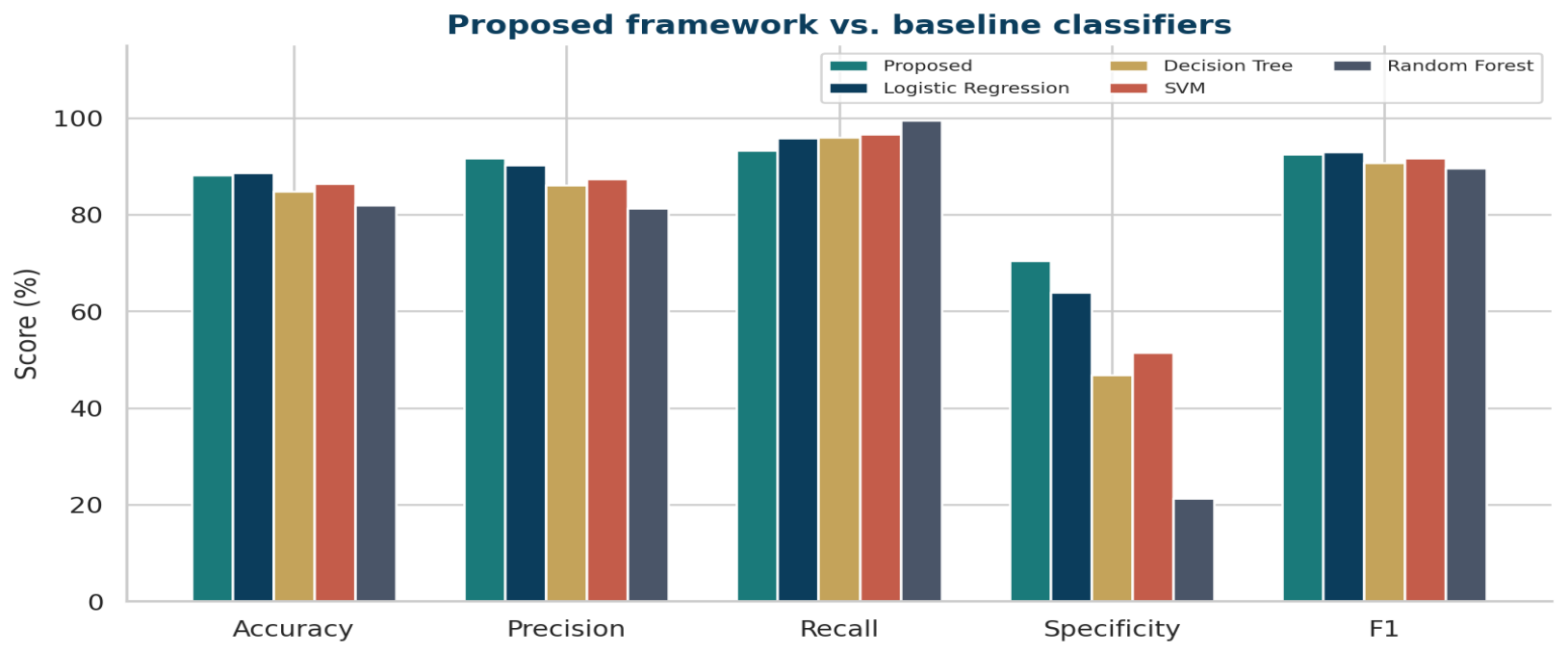
- 80/20 stratified split first.
- Mean impute + Min–Max, train-fit only.
- ANOVA SelectKBest, k = 30.
- SMOTE on the training set only.
- GMM K = 4; append 4 probabilities.
- XGBoost: 80 trees, depth 3 (no search).
- Baselines: LR, DT, SVM, RF — same split, no SMOTE/GMM.
- Seed 42.

Raw CSV | Impute / scale | SelectKBest | SMOTE | GMM K=4 | XGBoost | Test metrics

4. Results

88.1%	91.6%	93.3%	70.4%	92.4%	0.91
Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC





How to read these numbers (and why they are below the paper)

Paper reported Acc 93%, Recall 97%, F1 95% after heavy XGBoost tuning. Our run uses a small booster (80 trees, depth 3), 10% missing symptoms, and 6% self-report flips — closer to incomplete clinical forms. Recall stays high so true BD cases are still caught; specificity is lower, which is acceptable for a screening aid.

5. Conclusion, limitations, future work

Conclusion

- Three clear steps: SMOTE balances the train set, GMM adds 4 subgroup probabilities, XGBoost classifies.
- SMOTE and GMM never touch the test set, so the score is a fair hold-out estimate.
- Basic run: accuracy 88.1%, recall 93.3%, F1 92.4% — below the paper's 93 / 97 / 95 because we did not grid-search.
- High recall is the screening goal: better to flag a case than to miss BD.
- Feature importance from XGBoost keeps the model explainable.

Limitations

- One public Kaggle table, not a hospital cohort.
- BD labels follow a DSM-5 mania-count rule, so we added missing values and symptom noise to mimic incomplete forms.
- GMM assumes Gaussian clusters.
- SMOTE can invent samples near the class border.
- No EEG, speech, or imaging.
- We used a small XGBoost (80 trees, depth 3), so we do not claim the paper's 93%.

Future work

- Multimodal inputs: EEG, speech, actigraphy, neuroimaging.
- SHAP / LIME explanations for each flagged patient.
- Prospective validation across hospitals and age groups.
- Federated learning so sites share models, not notes.
- Longitudinal episode-risk monitoring.
- Calibrate the 0.5 threshold to local screening vs. confirmation workflows.